

Use-Case Flow Specification

Lab 1 — Requirements Engineering & UML Use-Case Modelling | Problem #02: Automated Rubric Assignment Evaluator

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Use Case Name: Generate Rubric Score Breakdown

Primary Actor: Faculty Evaluator (system-triggered after a student submission)

Supporting Actors: Student, Test Execution Engine

Preconditions

- The student is enrolled in the course and the submission window is open.
- The student has uploaded a valid project ZIP file that has passed the basic structure check.
- The faculty evaluator has already set the rubric criteria and unit tests.

Postconditions

- The system creates and saves the rubric score breakdown for the student's submission.
- The student and faculty evaluator can view the score breakdown.
- If evaluation fails, no incomplete score is saved; the submission is sent for manual checking and the student is notified.

Main Success Scenario

1. The student uploads the project ZIP file.
2. The system adds the submission to the processing queue.
3. The Test Execution Engine takes the submission and runs the configured unit tests in a separate environment.
4. The Test Execution Engine sends the test results and logs back to the system.
5. The system uses the test results to calculate the score for each rubric criterion.
6. The system saves the score with the student's submission details.
7. The score breakdown is shown to the student and faculty evaluator within 10 seconds, as required by FR-001.
8. The use case ends successfully.

Alternate Flow A1 — Faculty Changes an Automated Score

- A1.1. The faculty evaluator reviews the generated score.
- A1.2. If a score is incorrect or incomplete, the evaluator selects the criterion that needs to be changed.
- A1.3. The evaluator enters the new score and gives a short reason.
- A1.4. The system checks that the new score is within the allowed range.
- A1.5. The system saves the changed score while keeping the original automated score.
- A1.6. The updated score is shown to the student and faculty evaluator.
- A1.7. The use case ends successfully.

Related requirements: FR-001, FR-002, FR-004, NFR-001, NFR-002.

Related use cases: Submit Project ZIP, Execute Test Suite, Override Rubric Score.